respect to any k tuple $(\vec{x}_{i_1}, \ldots, \vec{x}_{i_k})$ as:

$$\mathsf{AD}_X(i, (i_1, \ldots, i_k)) = -\left(\frac{1}{k}\sum_{j=1}^{k} \log \frac{d(\vec{x}_i, \vec{x}_{i_j})}{d(\vec{x}_i, \vec{x}_{i_k})}\right)^{-1}. \tag{3}$$

The measure lid is then defined as: $\mathsf{lid}(i, X) = \mathsf{AD}_X(i, kNN_X(i))$.

Levina & Bickel (2004) defined the intrinsic dimensionality of a space as the average LID of all its points. This was later corrected by MacKay & Ghahramani (2005) to be the reciprocal of the average reciprocal of LID of each point. Formally defined as:

$$\mathsf{IntrDim}(X) = \left(\frac{1}{|X|}\sum_{\vec{x}_i \in X} \mathsf{lid}(i, X)^{-1}\right)^{-1}. \tag{4}$$

MacKay & Ghahramani (2005) showed that performing aggregation operations on the reciprocals of the LIDs of each point yields much better approximations.

3.1.2 Definition

To define Local NSA (LNSA) using LID, we start with the premise that closely structured point clouds should have similar dimensional manifolds in their local neighborhoods. Formally, for two point clouds X and Y, and considering the k nearest neighbours of the ith point in X indexed as $i_1, \ldots, i_k$, we examine the manifold dimensions in Y that contain $i, i_1, \ldots, i_k$. This leads to the concept of "Symmetric lid" defined as:

$$\overline{\mathsf{lid}}_X(i, Y) = \mathsf{AD}_Y(i, kNN_X(i)). \tag{5}$$

We notice that the mean discrepancy of the reciprocals of the LID is strongly correlated with the local structure being preserved, this is similar to MacKay & Ghahramani (2005) where they show that taking a mean of reciprocals of the LID is correlated to the intrinsic dimensionality of the whole space. Hence we define LNSA of X with respect to Y as:

$$\mathsf{LNSA}_Y(X) = N^{-1} \sum_{1 \le i \le N} \left(\mathsf{lid}(i, Y)^{-1} - \overline{\mathsf{lid}}_Y(i, X)^{-1}\right)^2. \tag{6}$$

Further, we can symmetricise this term to get the premetric $\mathsf{LNSA}^{\mathsf{metric}}$ as:

$$\mathsf{LNSA}^{\mathsf{metric}}(X, Y) = \mathsf{LNSA}_X(Y) + \mathsf{LNSA}_Y(X). \tag{7}$$

When using NSA-based loss functions, we use LNSA whereas when using NSA-based similarity indices, we use $\mathsf{LNSA}^{\mathsf{metric}}$.

3.1.3 Analytical Properties

We show that $\mathsf{LNSA}^{\mathsf{metric}}$ is a premetric by proving the necessary properties in Appendix D: • $\mathsf{LNSA}^{\mathsf{metric}}(X, X) = 0$, • Symmetry, • Non-negativity.

LNSA is shown to satisfy the necessary conditions of Invariance to Isotropic Scaling, Invariance to Orthogonal Transformation, and (Not) Invariance to Invertible Linear Transformation as proposed by Kornblith et al. (2019) for a similarity index. These condition are proved in Appendix E and ensure that the similarity index is unaffected by rotation or scaling of the representation space.

3.2 Global Normalized Space Alignment (GNSA)

GlobalNSA is a similarity index based on Representational Similarity Matrices (RSMs) (Klabunde et al., 2023). Given a representation $\mathbf{R} : N \times D$, all RSM based measures generate a matrix of instance wise similarities $D \in \mathbf{R}^{N \times N}$ where $D_{i,j} := d(R_i, R_j)$. Given two representations $\mathbf{R}$ and $\mathbf{R}'$, two corresponding RSMs are generated. GlobalNSA is calculated by subtracting the pairwise similarities between the two spaces and averaging these discrepancies to produce a single value that reflects the global difference between the two representation spaces. We present the motivation behind GlobalNSA and how RSMs capture global discrepancy in Appendix C.3.

3.2.1 Definition

The GNSA between two point clouds $X = \{\vec{x_1}, \ldots, \vec{x_N}\}$ and $Y = \{\vec{y_1}, \ldots, \vec{y_N}\}$ is computed as:

$$\mathsf{GNSA}(X, Y, i) = \frac{1}{N} \sum_{1 \leq j \leq N} \left| \frac{d(\vec{x_i}, \vec{x_j})}{\max_{\vec{x} \in X} d(\vec{x}, \vec{0})} - \frac{d(\vec{y_i}, \vec{y_j})}{\max_{\vec{y} \in Y} d(\vec{y}, \vec{0})} \right|.^{1} \tag{8}$$

$$\mathsf{GNSA}(X, Y) = \frac{1}{N} \sum_{1 \leq i \leq N} \mathsf{GNSA}(X, Y, i). \tag{9}$$

Throughout the paper, we use $d(\cdot, \cdot)$ to denote the euclidean distance (unless specified otherwise).

3.2.2 Analytical Properties

We show that GNSA is a pseudometric by proving the necessary properties in the Appendix A: • $\mathsf{GNSA}(X, X) = 0$, • Symmetry, • Non-negativity, • Triangle Inequality.
GNSA is proven to be a similarity index by satisfying the three established criteria in Appendix B.

3.3 Putting it all together

We utilize GNSA for our empirical analysis as it performs well when comparing entire representation spaces. When using NSA as a structure-preserving loss function, we calculate the final NSA Loss through a weighted sum of local and global NSA, allowing us to adjust the emphasis between local and global perspectives. One could prioritize local neighborhood preservation, such as in classifiers with tightly clustered data, or emphasize global structure, as needed in link prediction models for densely interconnected graphs. Hence, NSA is defined as:

$$\mathsf{NSA} = l * \mathsf{LNSA} + g * \mathsf{GNSA}. \tag{10}$$

Only GNSA is used during empirical analysis while both GNSA and LNSA are used during dimensionality reduction. We present ablation studies on how the two components affect performance, quantifying the difference and visualizing it in Appendix P.

3.3.1 Complexity Analysis

LNSA's computational complexity is $O(N^2D + kND)$ [2] and GNSA's complexity is $O(N^2D)$ where N is the number of data points, k is the number of nearest neighbours considered in LNSA and D is $\max(D(R), D(R'))$ with R and R' being the two representation spaces and $D(\cdot)$ the dimensionality of the space. Hence, overall NSA has complexity of $O(N^2D + kND)$. Empirically, we find that computing NSA on a GPU scales quadratically up to a very high batch size (Appendix H). CKA has similar complexity to NSA while RTD has a complexity of $O(N^2D + R^3)$, where R is the number of simplices (in practice $R >> N$). Running times of NSA-AE are given in Appendix K. NSA-AE is several times faster than RTD-AE and it can run on much larger batch sizes.

4 Applications

NSA serves a dual purpose, acting not only as a similarity index to compare structural discrepancies in representation spaces, but also as a structure preserving loss function. This enables a wide range of applications where NSA could potentially be useful. We first validate NSA's viability as a similarity index using the tests proposed by (Ding et al., 2021). We then present experiments in two broad domains; dimensionality reduction and adversarial attack analysis. Additional experiments on cross-layer analysis (Appendix T and Appendix I), convergence prediction (Appendix U) and manifold approximation (Appendix S) are presented in the appendix.

[1] $\vec{0}$ denotes a vector (of appropriate size) which is the origin of the space.

[2] The time complexity of computing the k-nearest neighbours of each point is $O(N^2D + N^2 \log k)$ since it takes $O(N^2D)$ time to find all pairwise distances and finding k minimum elements of N lists of size N can be done in $O(N^2 \log k)$ using min-heaps of size k. Since $D >> \log k$ in most cases, we write the total time complexity as $O(N^2D)$.

4.1 Analyzing Representation Spaces

We evaluate the performance of our structural similarity index using statistical tests proposed by Kornblith et al. (2019) and further refined by Ding et al. (2021). These tests are based on two fundamental intuitions: (1) a good structural similarity index should show high similarity between architecturally identical neural networks with different weight initializations, and (2) it should be sensitive to the removal of high variance principal components from the data. These initial tests are essential for validating the effectiveness of a similarity index.

4.1.1 Specificity Test

A structural similarity index is expected to assign small distances between corresponding layers of neural networks that differ only in their initialization seed. To test this, we trained two ResNet-18 models on the CIFAR-10 dataset, each with a different weight initialization. After training, we extracted intermediate embeddings from each layer during an evaluation pass over the entire dataset. Using a subset of data points[3], we computed the similarity between these embeddings using NSA, CKA, and RTD.

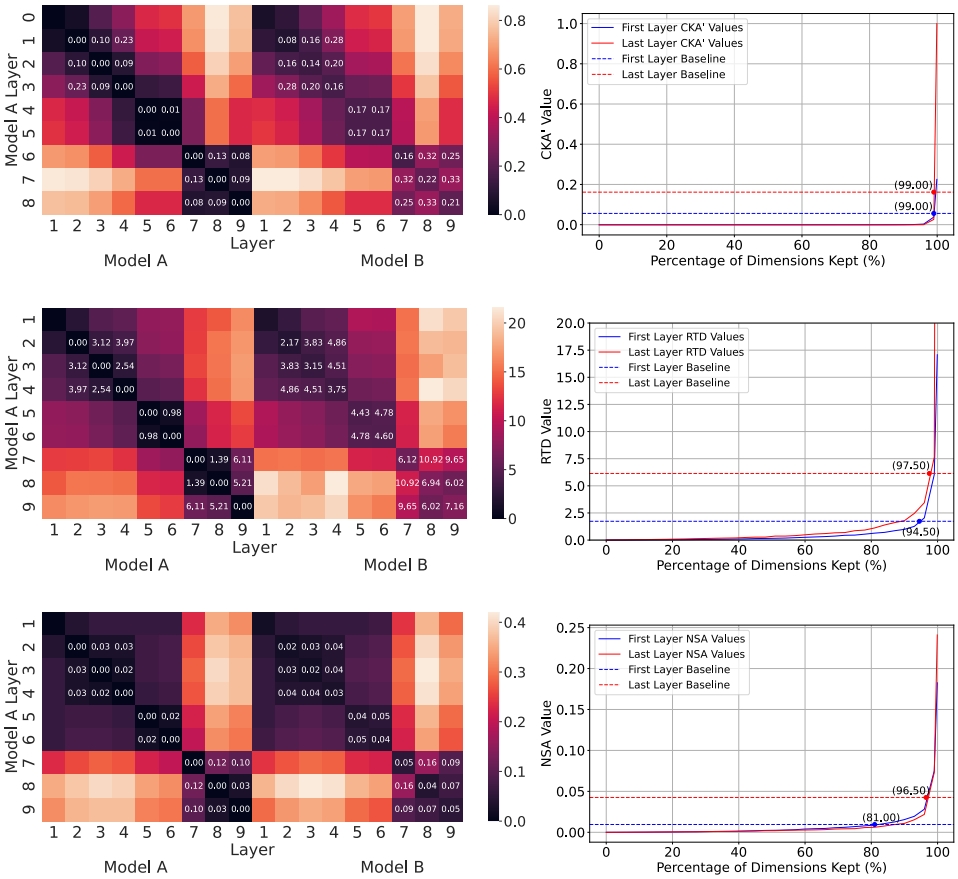

Figure 1: Specificity (Left) and Sensitivity (Right) Tests. Left (from top to bottom): CKA', RTD and NSA pairwise distances between each layer of two differently initialized networks. Right: Dissimilarities between a layer's representation and its low-rank approximation. Principal components are deleted in order of least variance.

Our results indicate that NSA exhibits the smallest drop in similarity between corresponding layers of networks with different initializations. Both RTD and CKA show a significant decrease in similarity, particularly in the later layers. NSA, on the other hand, maintains a strong layer-wise pattern, closely

[3]We set the subset size to 4000 for NSA and use the recommended subset size of 400 over 10 trials for RTD